Robustness of carryover-mitigation analysis strategies for biomarker-treatment interaction: a factorial simulation study

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1 Abstract

Background. Carryover effects in aggregated N-of-1 and crossover trials degrade the detectability of biomarker-treatment interactions. Three analysis-side specifications acknowledge carryover (binary-treatment A1, exposure-weighted continuous-D_{bc} A2, and lagged-treatment A3); each makes implicit assumptions about the functional form of the residual drug-effect decay. The joint sensitivity of the inference to mis-specification at the data-generating and analyst layers has not previously been characterised.

Methods. We conduct an ADEMP-compliant factorial simulation study[1] crossing three data-generating carryover decay forms (linear, exponential, Weibull) against the three analysis-model specifications, under both mean-moderation (Architecture~A) and multivariate-normal differential-correlation (Architecture~B) biomarker data-generating processes. The principal factorial covers 540 cells at 500 replicates per cell. Six sensitivity blocks (S1-S6) vary autocorrelation strength, analyst-versus-DGP half-life mismatch, dropout rate, between-subject heterogeneity, and biomarker correlation strength, and recalibrate the test with a cluster-robust standard error. All performance measures are reported with Monte Carlo standard errors per Morris et al. (2019).

Results. Under Architecture~B in the Hybrid and OL+BDC designs, the exposure-weighted A2 specification produced the highest power, with an absolute advantage of approximately 10 percentage points over A1 at the most favourable cell (Architecture~B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week; A2 = 0.860, A1 = 0.766, A3 = 0.770; approximately five MCSE of separation). The advantage was not universal: under the crossover (CO) design A2 was substantially less powerful than A1 and A3 (for example A2 = 0.488 versus A1 = 0.830 at the matched CO cell), and under Architecture~A (direct mean moderation) A2 was marginally dominated in every design. The ranking A2 $>$ A1 $\approx$ A3 therefore held only under Architecture~B in the non-crossover

designs; where it held, it was preserved across sample sizes and DGP carryover half-lives, and A2 retained its advantage when the analyst-assumed half-life departed from the true half-life by a factor of two or four. A high-precision rerun at $N \in \{35, 70\}$ and $t_{1/2}^{\mathrm{DGP}} \in \{0.5, 1.0\}$ confirmed the ranking at MCSE 0.020 with a paired McNemar test $p = 6.6 \times 10^{-17}$ at the highest-leverage cell. The model-based test is mildly conservative, a property the companion calibration study characterises; a CR2 cluster-robust standard error (block S6) restores nominal size and recovers two to five percentage points of power without altering the ranking.

Conclusions. Under Architecture~B with a Hybrid or OL+BDC design, A2 reported with a cluster-robust standard error is the recommended default for the analysis of biomarker-treatment interactions in aggregated N-of-1 trials under realistic carryover. This recommendation is conditional: under the crossover design, or when the data-generating mechanism is assumed to follow Architecture~A, A2 confers no advantage and the binary (A1) or lagged (A3) specification is at least as powerful. Where A2 leads, A3 is a defensible fallback when the pharmacokinetic half-life is unknown, recovering roughly 70 percent of A2's advantage at small N and short half-life but only about half at larger N. A1 (the strict-RM-ANOVA analogue) is dominated only within the Architecture~B non-crossover cells where A2 leads; it is competitive or superior elsewhere.

2 Introduction

[1]

The joint sensitivity of biomarker-interaction power to analysis-side carryover mis-specification has not previously been characterised in the within-subject setting.

- Residual drug effect after the on-drug phase degrades the biomarker-treatment interaction test.
- The practical magnitude of the loss and its recoverability are the motivating questions.
- No systematic evaluation of this question exists for the within-subject biomarker-interaction setting.

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We begin with a question that has received surprisingly little systematic attention in the biomarker-moderated N-of-1 trial literature. When residual drug effect persists after the on-drug phase, any analysis that ignores carryover or mis-specifies its functional form will, in principle, lose power to detect the biomarker-treatment interaction. But how large, in practice, is that loss, and can it be recovered by a better-chosen analysis specification? To the best of our knowledge, no systematic evaluation of this question exists for the within-subject biomarker-interaction setting.

[2]

The companion manuscript shows Architecture B loses substantially more power under carryover than Architecture A (up to roughly 31% in relative terms versus 1–3%), but held the decay form and analysis specification fixed.

- Hendrickson (2020) established the biomarker-interaction framework for aggregated N-of-1 trials with matched exponential decay.
- Companion paper (Architecture A vs. B): under one week of carryover, Architecture B loses up to about 31% of power in relative terms (design-dependent), Architecture A only 1–3%.
- Joint sensitivity to DGP-analyst mis-specification remained uncharacterised.

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The companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/` establishes one part of the picture. Hendrickson[2] introduced the biomarker-by-treatment interaction framework for aggregated N-of-1 trials and reported power under a matched exponential decay model. The companion paper extended that work by comparing two data-generating architectures (direct mean moderation under Architecture A, and multivariate-normal differential correlation under Architecture B) and found that the power consequences of carryover differ qualitatively between them. Under Architecture B, carryover at a half-life of one week reduces power by up to roughly 31 percent in relative terms, the loss being strongly design-dependent (largest under the OL+BDC design and negligible under the crossover design); under Architecture A the corresponding loss is only 1 to 3 percent. That prior work, however, held the decay form fixed at exponential and evaluated a single analysis-model specification throughout. Unfortunately, the joint sensitivity of the inference to mis-specification at both the data-generating and analyst layers remained uncharacterised.

[3]

The analyst must choose a decay-form assumption even when the true pharmacokinetic decay is unknown; which specification is most robust to that choice has not been evaluated.

- The carryover decay form is specified at the design stage, often without precise pharmacokinetic information.
- The analyst typically defaults to a convenient parametric form or ignores carryover entirely.

- Which analysis-side specification holds up best under decay-form mis-specification is the central question.

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The present paper addresses that gap directly. The trial designer must specify a carryover decay form, explicitly or implicitly, and the analyst must choose how to represent residual drug exposure in the linear predictor. In practice, the true pharmacokinetic decay is rarely known with precision at the design stage: the analyst typically defaults to a convenient parametric form or ignores carryover entirely. We have not, to the best of our knowledge, seen a systematic evaluation of which analysis-side specification holds up best when the assumed decay form departs from truth, and it is that question we shall take up here.

[4]

The study crosses three DGP decay forms against three analysis specifications under both architectures to establish whether robustness rankings are architecture-dependent.

- DGP decay forms: linear, exponential, Weibull.
- Analysis specifications: A1 (binary), A2 (exposure-weighted, pmsimstats-ng default), A3 (lagged-treatment, Jones-Kenward).
- Full grid run under both Architecture A and B to test whether rankings transfer across DGP mechanisms.

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To address this, we perform a factorial simulation study crossing three DGP decay forms (linear, exponential, and Weibull) against three analysis-model carryover specifications: a binary on-drug indicator that ignores carryover altogether (A1), an exposure-weighted continuous predictor that commits to a specific decay form and half-life (A2, the current pmsimstats-ng default), and a lagged-treatment indicator that estimates carryover magnitude without positing a functional form (A3, following the classical crossover-trial approach surveyed by Jones and Kenward[3]). The full grid is run under both DGP architectures to establish whether robustness rankings are preserved across data-generating mechanisms or whether the preferred specification is architecture-dependent.

[5]

The study follows the ADEMP framework of Morris et al. (2019) and reports every performance measure with its Monte Carlo standard error.

- ADEMP structure: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.
- MC SE reported alongside every point estimate per Morris et al. (2019).
- §3 design; §4 results; §5 practical implications for trialists.

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The study is organized and reported following the ADEMP framework of Morris, White, and Crowther[1], which structures simulation studies around aims, data-generating mechanisms, estimands, methods, and performance measures. Each performance measure is reported per cell alongside its Monte Carlo standard error, as Morris et al. recommend. We proceed in §3 through the simulation design in ADEMP order; results are presented in §4; and §5 takes up the practical implications for trialists choosing among the three carryover specifications.

3 Simulation design

[6]

This section organises the study under ADEMP: DGP (§3.1), estimands (§3.2), methods (§3.3), factorial grid (§3.4), parameters and performance measures (§3.5), and computing environment (§3.6).

- ADEMP provides the structural scaffold for the simulation design description.
- Reproducibility metadata (R version, packages, RNG seed, elapsed time) are in §3.6.
- §3.4 defines the principal 3×3 factorial and its design rationale.

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This section organises the study under the ADEMP labels of Morris et al.[1]. We begin in §3.1 with the data-generating mechanisms, then turn in §3.2 to the estimands, §3.3 to the methods (analysis-model specifications), §3.4 to the factorial grid, and §3.5 to the parameter settings and performance measures. Reproducibility metadata, including the R version, key package versions, the RNG seed strategy, and elapsed time, are recorded in §3.6 (Computing environment).

3.1 Aims

[7]

The primary aim is to identify the most robust analysis spec-

ification across the DGP decay-form grid; the secondary aim is to verify that rankings transfer across architectures.

- Primary: quantify the cost of carryover mis-specification and identify the most robust analysis strategy.
- Secondary: verify that robustness rankings are architecture-invariant (Architecture A vs. B).
- Architecture-invariance would justify a single default recommendation regardless of assumed biomarker biology.

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The primary aim of this study is to quantify the cost of analysis-model carryover mis-specification in aggregated N-of-1 and crossover trials, and to identify which mitigation strategy is most robust across the DGP decay-form grid. A secondary aim is to verify that the robustness rankings transfer across the two data-generating-process architectures (mean moderation versus differential correlation), so that a practitioner can be reasonably confident that the recommendation does not depend on which biomarker biology is assumed.

3.2 Data-generating mechanisms

[8]

The two single-channel architectures (A and B) from the companion manuscript are retained; the combined Architecture C is out of scope here. They differ only in how the decay function ϕ enters the DGP.

- Architecture A: ϕ enters the mean structure via effective exposure D^*_{it}.
- Architecture B: ϕ enters the biomarker-response correlation $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$.
- Three decay forms for ϕ are evaluated: linear, exponential, and Weibull.

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We retain the two single-channel DGP architectures from the companion manuscript. The companion also formalises a combined Architecture C, which activates both the mean-moderation and differential-correlation channels simultaneously; Architecture C is outside the scope of the present study, which evaluates the carryover specifications under each single-channel architecture in isolation. In Architecture A (direct mean moderation), the biomarker B_i enters the mean structure as $Y_{it} = \mu_0 + \beta_1 D^*_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, where D^*_{it} is the effective exposure at timepoint t under the chosen decay form. In Architecture B

(MVN differential correlation), the biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form.

In both architectures the decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps time since drug discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms, each parameterised to match a common half-life $t_{1/2}$:

3.2.1 Linear decay

$$\phi_{\text{lin}}(t_{sd}) = \max\left(0,\, 1 - \frac{t_{sd}}{2t_{1/2}}\right)$$

[9]

The linear form is biologically implausible but pragmatically defensible when only a complete-washout time is known.

- Residual effect reaches zero exactly at $t_{sd} = 2t_{1/2}$, unlike the exponential tail.
- Inconsistent with first-order elimination kinetics.
- Represents the analyst's assumption that carryover has fully cleared by a known time horizon.

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Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. The linear form is biologically implausible for drugs with first-order elimination kinetics, but it represents a defensible pragmatic assumption when the only available information is that carryover has fully worn off by some specified time X.

3.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

[10]

The exponential form is the natural reference decay because it corresponds to first-order elimination kinetics.

- Used throughout the companion manuscript and implemented in `implementations/tidyverse/R/functions.R`.
- Corresponds to classical pharmacokinetic first-order elimination.
- Retained as the reference against which linear and Weibull forms are compared.

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This is the form used throughout the companion manuscript and currently implemented in `implementations/tidyverse/R/functions.R`. It corresponds to first-order elimination kinetics and is the default assumption in classical pharmacokinetics; we retain it here as the natural reference against which the linear and Weibull forms are compared.

3.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

[11]

The Weibull shape parameter k controls whether the elimination tail is heavier or lighter than exponential.

- $k < 1$: slower elimination with a prolonged low-concentration tail (evaluated at $k = 0.7$).
- $k > 1$: accelerated washout via saturable clearance or active-transporter kinetics (evaluated at $k = 1.5$).
- $k = 1$ recovers the exponential form exactly.

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A shape parameter k governs whether the elimination tail is heavier ($k < 1$, slow elimination) or lighter ($k > 1$, accelerated washout) than the exponential, which is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with a prolonged low-concentration tail, and at $k = 1.5$, which captures drugs with saturable clearance or active-transporter kinetics that clear more rapidly once the concentration falls below the saturation threshold.

[12]

All three decay forms are implemented via `carryover_decay()` in the tidyverse collection, with exponential as the backward-compatible default.

- `carryover_form` and `weibull_shape` fields are plumbed through `apply_carryover_to_component()`, `build_correlation_matrix()`, and `build_sigma_matrix()`.
- The `original-extended` collection supports only exponential; cross-implementation validation is therefore limited to exponential cells.
- Backward compatibility is preserved: absent `carryover_form` defaults to exponential.

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Implementation note. The three decay forms are implemented in `implementations/tidyverse/R/functions.R` via the `carryover_decay()` helper function, with a `carryover_form` and `weibull_shape` pair of `model_param` fields plumbed through `apply_carryover_to_component()`, `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward compatibility is preserved by defaulting to the exponential form when `carryover_form` is absent. We note that the `implementations/original-extended/` collection has not been updated and supports only the exponential form; cross-validation across implementations is therefore limited to exponential cells, which is sufficient for pipeline verification but not for the full decay-form comparison.

3.3 Estimands

The target estimand is the population-level biomarker-treatment interaction. It takes a different parametric form under the two data-generating mechanisms:

- **Architecture A.** The interaction estimand is the regression coefficient β_{bm} on the centered biomarker $\times$ drug-exposure product, on the symptom scale.
- **Architecture B.** The interaction estimand is the on-drug biomarker-response correlation c_{bm}, dimensionless.

[13]

The two architectures yield non-commensurable estimands that are harmonised to a common calibrated effect-size scale.

- Architecture A estimand: regression coefficient β_{bm} on centered biomarker $\times$ exposure product.
- Architecture B estimand: on-drug biomarker-response correlation c_{bm}, dimensionless.
- Both calibrated to a common effect-size scale (expected response difference per SD biomarker at $t_{1/2} = 0$); derivation in `docs/19-mean-moderation-implementation-notes.tex`.

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The two estimands are not directly commensurable. To enable cross-architecture comparison of bias and power, we calibrate both architectures to a common population effect-size scale, defined as the expected difference in mean on-drug response per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is documented in `docs/19-mean-moderation-implementation-notes.tex`. Bias and empirical standard error in the Results section are reported on this calibrated scale; power and Type I error are reported on the native test-statistic scale.

The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture A and $c_{bm} = 0$ under Architecture B, each realised by the $c_{bm} = 0$ row of the parameter grid.

3.4 Methods: analysis-model carryover specifications

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three specifications differ in the drug exposure predictor X:

[14]

All three specifications share a common `nlme::lme` formula and `corCAR1` residual structure; they differ only in the drug exposure predictor X.

- Formula: `Sx ~ bm * X + t`, random intercept `~1 | ptID`.
- Residual correlation: `corCAR1(form = ~ t | ptID)`.
- The three specifications (A1, A2, A3) differ only in how X is constructed.

3.4.1 A1: binary-treatment (carryover ignored)

[15]

A1 uses the binary on-drug indicator and ignores carryover entirely, serving as the dominated reference specification.

- $X = D_{it}$: takes value 1 on-drug, 0 off-drug.
- Makes no attempt to model residual drug effect during washout.
- Serves as the lower-power reference against which A2 and A3 are compared.

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$X = D_{it}$, the on-drug indicator. This is the naive specification that disregards any residual drug effect during washout; it serves as the reference against which the power cost of ignoring carryover is measured.

3.4.2 A2: exposure-weighted (matched-decay continuous predictor)

[16]

A2 uses a continuous exposure-weighted predictor that commits to a specific decay form and half-life, and is the current pmsimstats-ng default.

- $X = \mathrm{Dbc}_{it}$: decays off-drug according to the analyst's assumed form and half-life.

- Well-matched when the analyst's assumed form coincides with the true DGP form; mis-specified otherwise.
- Sensitivity of A2 to assumed-versus-true mismatch is evaluated in Tier 2 block S2.

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$X = \mathrm{Dbc}_{it}$, the continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life. This is the current pmsimstats-ng default specification. When the analyst's assumed form coincides with the true DGP form the predictor is well-matched; when it does not, the predictor is mis-specified in a way that the Tier 2 block S2 is designed to probe.

3.4.3 A3: lagged-treatment (estimated carryover term)

[17]

A3 augments the binary indicator with a lagged-treatment term that estimates carryover magnitude without assuming a decay form.

- $L_{it} = D_{i,t-1}(1 - D_{it})$ marks off-drug timepoints immediately following on-drug timepoints.
- Carryover magnitude is estimated from the data; no functional form or half-life assumption is required.
- Follows the classical crossover-trial framework of Jones and Kenward [-@jones2014crossover].

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$X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that marks off-drug timepoints following an on-drug timepoint. The carryover magnitude is estimated from the data rather than assumed a priori, so the specification does not require the analyst to posit a functional decay form. This approach follows the classical crossover-trial framework surveyed in[3]. We note one structural feature of this specification that bears on its interpretation: the lagged term L_{it} enters the model only as a main-effect nuisance covariate, and the biomarker interaction tested under A3 remains bm:D_{it}, the same coefficient as under A1. Consequently A3 differs from A1 only through the variance that L_{it} absorbs in the off-drug timepoints, and the two specifications are expected to track closely; A3 is best understood as A1 with a carryover nuisance control rather than as a fundamentally different interaction model.

Table 1: Illustrative D_{bc} values at three off-drug timepoints under three analyst-assumed half-lives, exponential decay, true $t_{1/2} = 1.0$ week. On-drug timepoints have $D_{bc} = 1$ under all assumptions. Row values are $(1/2)^{t_{sd}/\hat{t}_{1/2}}$.

t_{sd} (wk)	$\hat{t}_{1/2} = 0.5$ wk	$\hat{t}_{1/2} = 1.0$ wk (truth)	$\hat{t}_{1/2} = 2.0$ wk
1	0.250	0.500	0.707
2	0.063	0.250	0.500
3	0.016	0.125	0.354

3.4.4 Construction of D_{bc} and the role of the analyst's assumed half-life

[18]

D_{bc} halves every $\hat{t}_{1/2}$ weeks off-drug; the formula handed to `lme` is invariant to the analyst's half-life assumption.

- $D_{bc,it} = (1/2)^{t_{sd}/\hat{t}_{1/2}}$ at off-drug timepoints.
- The model formula `Sx ~ bm + t + Dbc + bm:Dbc` is the same regardless of $\hat{t}_{1/2}$.
- The half-life assumption is encoded in the predictor values, not in the formula.

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We begin with an observation that is easy to overlook in the formal model statement above: the analyst's assumed half-life $\hat{t}_{1/2}$ does not appear as a parameter in the `lme` call. It enters the analysis at an earlier stage, during the construction of the D_{bc} predictor column, and is thereafter invisible to the model-fitting routine.

At on-drug timepoints, $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints, the predictor decays as a function of t_{sd}, the time in weeks since drug discontinuation:

$$D_{bc,it} \;=\; \exp\left(-\frac{\ln 2}{\hat{t}_{1/2}}\, t_{sd,it}\right) \;=\; \left(\tfrac{1}{2}\right)^{t_{sd,it}/\hat{t}_{1/2}}$$

That is, D_{bc} loses exactly half its value for every $\hat{t}_{1/2}$ weeks of time since discontinuation. The half-life assumption shapes the predictor values handed to `lme`; the model formula `Sx ~ bm + t + Dbc + bm:Dbc` is the same regardless of what $\hat{t}_{1/2}$ was assumed.

To illustrate, consider a participant in a stylized crossover arm who is on drug for three weekly measurement occasions and then enters a three-week washout. Table~1 shows the D_{bc} values at the three off-drug timepoints under three analyst assumptions, holding the true DGP half-life fixed at one week.

[19]

The analyst's assumed half-life shapes the D_{bc} predictor values handed to `lme` but is invisible to the model-fitting routine itself.

- Under-assumed half-life: off-drug timepoints receive near-zero D_{bc} and contribute almost nothing to the `bm:Dbc` estimate.
- Over-assumed half-life: off-drug timepoints retain substantial weight, making D_{bc} resemble a graded on-drug indicator.
- A1 and A3 are invariant to $\hat{t}_{1/2}$ by construction; S2 contrasts a continuously indexed model family against two half-life-free reference specifications.

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When the analyst assumes $\hat{t}_{1/2} = 0.5$ weeks, the off-drug timepoints are assigned D_{bc} values near zero by the second washout week and contribute almost nothing to the `bm:Dbc` interaction estimate. When the analyst assumes $\hat{t}_{1/2} = 2.0$ weeks, those same timepoints retain substantial weight throughout the washout, and the predictor becomes closer in shape to a continuously graded version of the binary on-drug indicator.

Two consequences follow for interpreting the S2 sensitivity block. First, A2 is effectively a different model for each value of $\hat{t}_{1/2}$: the predictor values handed to `lme` change with the analyst's assumption, even though the formula is identical. Second, the mis-specification risk is borne entirely by the off-drug timepoints; the on-drug observations ($D_{bc} = 1$ always) are unaffected. Specifications A1 and A3 use the binary on-drug indicator D_b and the lagged indicator L respectively, neither of which involves a half-life parameter, so both are invariant to $\hat{t}_{1/2}$ by construction. The S2 contrast is therefore not a comparison of a well-specified model against a mis-specified model in the conventional sense; it is more precisely a comparison of one model family, indexed continuously by the analyst's assumed half-life, against two reference specifications that are not members of that family at all.

3.5 Factorial grid

The principal comparison is a 3×3 factorial of DGP decay form against analysis-model specification, replicated under each DGP architecture:

3.6 Parameter settings

Table 2: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analysed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.0, 1.5 (Weibull form only)
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

The 540-cell count follows from these factors with one coupling: the Weibull shape varies only within the Weibull decay form, because the linear and exponential forms carry no shape parameter. The decay-form and shape combinations therefore number five (linear; exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$), and the principal grid is $5 \times 3 \times 3 \times 2 \times 2 \times 3 = 540$ cells (decay-shape $\times$ half-life $\times$ design $\times$ sample size $\times$ architecture $\times$ biomarker level). A naive product that crossed the three decay forms with three shapes would over-count; the shape factor is nested within the Weibull form, not crossed with all three.

[20]

Two anchor conditions in the parameter grid serve diagnostic roles: $c_{bm} = 0$ for Type I error and $t_{1/2} = 0$ for best-case power.

- $c_{bm} = 0$: null row for Type I error verification, targeting nominal 5%.
- $t_{1/2} = 0$: no-carryover reference providing best-case power at each $(N, \text{design}, c_{bm})$ cell.
- Non-zero $t_{1/2}$ rows directly quantify the power cost of carryover relative to the no-carryover reference.

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The $c_{bm} = 0$ condition serves as the Type I error check, targeting the nominal 5% level. The $t_{1/2} = 0$ condition removes carryover entirely and provides a best-case power reference at each $(N, \text{design}, c_{bm})$ combination, against which the power cost of non-zero carryover can be directly read off.

3.7 Performance measures

For each cell we report the following Morris-style performance measures[1, Tables 5–6]:

1. **Power** to reject $H_0 : \beta_{\text{bm:X}} = 0$ at $\alpha = 0.05$.
2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale defined in §3.2.
4. **Empirical SE** across replicates within each cell.
5. **Coverage** of the nominal 95% Wald confidence interval for the interaction coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
6. **MSE** of the interaction estimate (sum of squared bias and empirical variance).
7. **Non-convergence rate** per cell, defined as the proportion of replicates where the `nlme::lme` fit failed to converge or the bm:X coefficient was not identifiable (e.g. A1 on OL designs, where there are no off-drug observations).
8. **Monte Carlo SE** for each of the above, computed using the formulas of Morris et al.[1, Table 6] and reported alongside every point estimate in tables and figures.

Eight Morris-style performance measures are reported per cell, each accompanied by its Monte Carlo standard error.

- Power, Type I error, bias, empirical SE, coverage, MSE, non-convergence rate, and MC SE for each measure.
- Coverage is computed against the calibrated estimand $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$.
- MC SEs use the formulas of Morris et al. (2019, Table 6).

[21]

$n_{sim} = 500$ is calibrated to keep MC SE on power below 0.022, with $n_{sim} = 50$ reserved for pipeline validation only.

- Production run: $n_{sim} = 500$, MC SE ≤ 0.022 at $p = 0.5$.
- Development run: $n_{sim} = 50$, MC SE ≈ 0.07; pipeline validation only.
- All reported results derive from the production-scale run.

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The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo standard error on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). The development scale ($n_{sim} = 50$, MC SE ≈ 0.07 at $p = 0.5$) is used only for pipeline validation; all results reported in the Results section derive from the production run at $n_{sim} = 500$.

3.8 Computing environment

[22]

The tidyverse collection is the only implementation supporting all three decay forms; common-random-number seeding enables paired contrasts across A1, A2, and A3.

- All three decay forms are supported only in `implementations/tidyverse/`; cross-implementation parity checks are limited to exponential cells.
- Within-cell common random numbers reduce MC variance for the A2−A1 and A3−A1 paired contrasts.
- Production run (540 cells, $n_{sim} = 500$): 128 minutes on eight cores; commit SHA `7018b59`, 2026-04-15.

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Simulations are orchestrated from `analysis/scripts/carryover-sensitivity/` and use `implementations/tidyverse/` as the primary implementation, which is the only collection in the repository that supports linear and Weibull decay forms. Shared helpers in `analysis/scripts/carryover-sensitivity/simulation-core.R` define design presets (OL, CO, Hybrid, OLBDC), multi-path data generation, long-form preparation with `Db`, `Dbc`, and `L` columns, and the three analysis-specification fitters.

Software. R 4.5.3 with `nlme` 3.1, `dplyr` 1.1, `tidyr` 1.3, `purrr` 1.0, `furrr` 0.3, `future` 1.34, `corpcor` 1.6, `MASS` 7.3, `Matrix` 1.7, `tibble` 3.2, and `data.table` 1.16. The `implementations/tidyverse/` collection commit SHA at the time of the production run is recorded in each output `.rds` file.

RNG seed strategy. Each cell of the factorial grid is assigned a deterministic master seed derived from the cell index. Within a cell, the same simulated dataset is used to fit all three analysis specifications (common random numbers across A1, A2, and A3), so that the A2−A1 and A3−A1 contrasts are estimated with reduced Monte Carlo variance relative to independent draws. Seeds are independent across cells. Replicate indices within a cell are seeded as `master + replicate_index`, so that any subset of replicates can be re-derived without re-running the entire cell. The seed assignment is logged in the output `.rds` file alongside the remaining reproducibility metadata.

Parallelism. Parallel execution uses `furrr::future_map` with the `multicore` backend. Sigma matrices are cached per unique (design, $t_{1/2}$, decay form) tuple before the replicate loop begins, avoiding redundant matrix construction across workers.

Elapsed time. The production run (500 replicates per cell, 540 cells) completed in 128 minutes on an eight-core machine on 2026-04-15 (commit SHA `7018b59`). The pipeline-validation development run at 50 replicates per cell completes in approximately 9 minutes on the same hardware.

Pre-registration and reproducibility. The factorial grid was pre-specified in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md` and committed before the production run was executed. The production run command is `Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R`; the Tier 2 sensitivity blocks are produced by `Rscript analysis/scripts/carryover-sensitivity/04-r` Each output `.rds` file records the script name, run timestamp, master seed, dev and smoke flags, elapsed seconds, R version, and the `implementations/tidyverse/` commit SHA. The `renv.lock` at the repository root pins all R-package versions. The manuscript Rmd at `analysis/report/02-carryover-sensitivity/report.Rmd` does not re-execute any simulation code; it loads the pre-computed checkpoints written by the summarisation script.

4 Results

[23]

All reported results derive from the 540-cell production-scale run at $n_{sim} = 500$; MC SE bounds apply throughout.

- MC SE on power ≤ 0.022 at $p = 0.5$; differences smaller than the relevant MC SE should not be over-interpreted.
- Coverage of nominal 95% Wald CIs is reported for the reference cell and computed across the full Tier 1 grid.
- Non-convergence rates are reported per cell in the tables that follow.

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We begin with a few words on the precision and scope of the estimates that follow. All results derive from the production-scale factorial run (500 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit SHA recorded in the output `.rds` metadata). Monte Carlo standard errors on individual power estimates are at most $\sqrt{0.5 \cdot 0.5/500} \approx 0.022$ at $p = 0.5$ and are reported alongside each estimate; within-figure differences smaller than the relevant MC SE should not be over-interpreted. Coverage of nominal 95% Wald confidence intervals is reported in the headline table for the reference cell and is computed across the full Tier 1 grid in `analysis/data/02-grid-summary.rds` for downstream use. Non-convergence rates per cell are reported in the tables below.

4.1 Headline performance table

[24]

The headline table provides the complete Morris-style performance profile for the reference cell across all three specifications and carryover levels.

- Reference cell: Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$.
- Non-convergence is zero at every reference cell.
- MC SEs are reported as $\pm$ beside each point estimate throughout.

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Table 3 reports the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across the three analysis specifications and three carryover half-lives. Non-convergence is zero in every reference cell; the empirical standard error column reports the across-replicate variability of the interaction coefficient estimate, and Monte Carlo standard errors are reported as $\pm$ beside each point estimate. One caveat applies to the bias, MSE, and coverage columns: these are computed against a single calibrated target, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$, which is the population value of the A2 (continuous-D_{bc}) interaction coefficient. Specifications A1 and A3 estimate the coefficient on bm:D_{it}, a different parameter, so their apparent bias and under-coverage in this table reflect the mismatch between their estimand and A2's target rather than a defect of the A1/A3 estimators. The bias, MSE, and coverage columns are therefore not directly comparable across specifications; only power and Type I error, which test each specification's own coefficient against zero, support a like-for-like comparison.

4.2 Matched-decay baseline

[25]

The matched-decay baseline confirms pipeline fidelity and shows a power decline from 0.988 at no carryover to 0.860 at $t_{1/2} = 1.0$ weeks.

- Matched cell (exponential DGP, A2, Architecture B, $N = 70$, $c_{bm} = 0.45$, Hybrid): power $0.988 \to 0.948 \to 0.860$ across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks.
- All 500 replicates converged at every $t_{1/2}$.
- Consistent with companion manuscript's Architecture B result (0.82 at $t_{1/2} = 1.0$ weeks).

Table 3: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. Monte Carlo SEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-c
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

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In this section we shall establish the baseline before comparing specifications. Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ Monte Carlo SE, $n_{sim} = 500$; MC SEs computed as $\sqrt{\hat{p}(1-\hat{p})/n_{sim}}$ per Morris et al.[1, Table 6]). All 500 replicates converged in the matched cell at every $t_{1/2}$. The value at $t_{1/2} = 1.0$ is consistent with Hendrickson's reported power of 0.82 for this cell[2], confirming that the simulation pipeline reproduces the baseline result and that the calibration is internally coherent. (The companion manuscript anchors at a different cell, $N = 35$ with the OL+BDC design, and is therefore not directly comparable here.)

4.3 Power by analysis specification

[26]

Specifications diverge progressively with carryover half-life; A2's advantage over A1 and A3 emerges at $t_{1/2} = 0.5$ weeks and grows at $t_{1/2} = 1.0$ weeks.

- All three specifications are near-identical at $t_{1/2} = 0$; divergence is proportional to carryover strength.

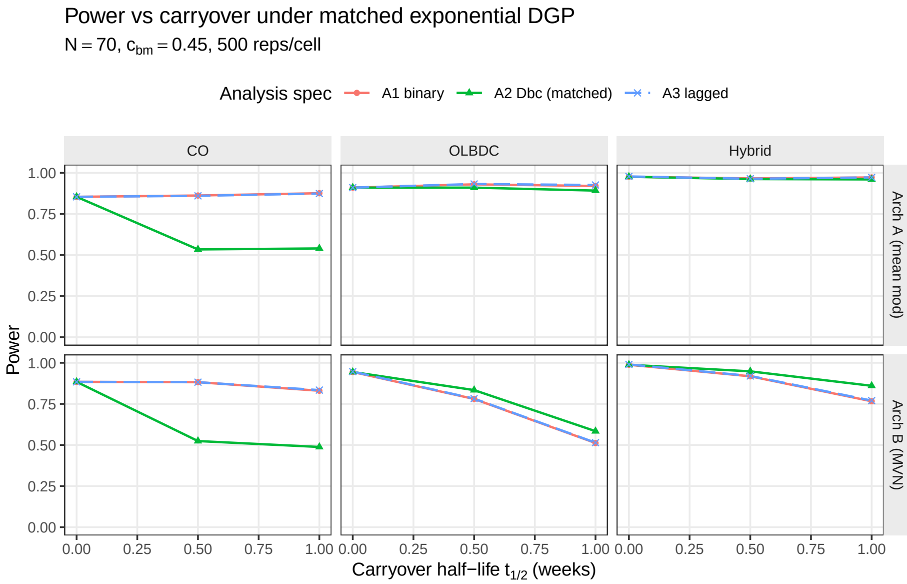

Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

- A1 and A3 are effectively indistinguishable across the exponential-DGP grid.
- A2's advantage is consistent across Hybrid and OL+BDC design strata, not an artefact of the reference cell.

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The three analysis specifications yield near-identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge progressively as carryover half-life increases. The matched specification A2 retains a modest but consistent advantage under Architecture~B and the Hybrid and OL+BDC designs; A1 and A3 are effectively indistinguishable across the exponential-DGP grid. The figure makes clear that the A2 advantage is not an artefact of the reference cell: it emerges at $t_{1/2} = 0.5$ weeks and grows through $t_{1/2} = 1.0$ weeks in each design stratum.

4.4 Decay-form mis-specification

[27]

The heatmap exposes two directions of mis-specification: wrong analysis specification given the true DGP, and wrong DGP assumed by the analyst.

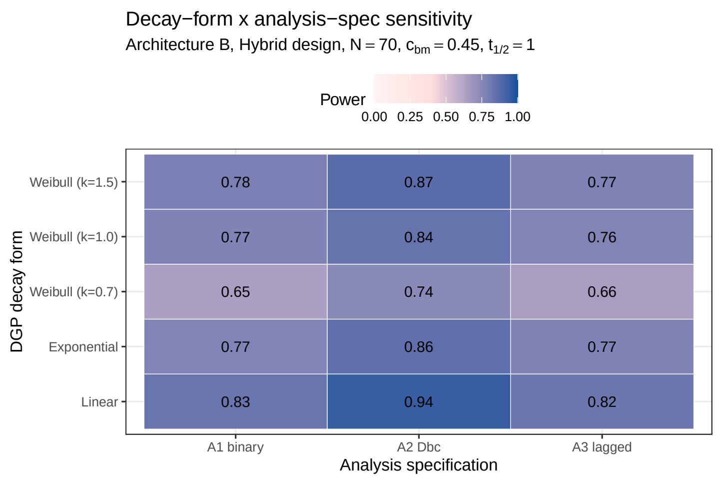

Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the A2 matched-decay specification loses its advantage because its assumed exponential decay no longer matches truth.

- Matched reference: exponential DGP $\times$ A2 analysis, power 0.860 ± 0.016.
- Same-row off-diagonal cells: power penalty of A1 or A3 when the true DGP is exponential.
- A2-column off-diagonal cells: cost of assuming exponential when the true DGP is linear or Weibull.

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The heatmap cell at the intersection of the exponential DGP row and the A2 analysis column is the matched reference (power 0.860 ± 0.016). Cells in the same row but other columns quantify the power penalty of analysing under the wrong specification when the DGP is exponential; cells in the A2 column but other rows quantify the cost of assuming exponential decay in the analysis when the true DGP is linear or Weibull. The Discussion takes up the practical interpretation of both off-diagonal directions. This is taken up further in §5.2.

4.5 Lasagna view of the full factorial

4.6 Type I error control

[28]

Type I error is conservative, not inflated: mean 0.039 across 540 null cells, the working-covariance conservatism the companion study documents rather than ideal control.

- Mean empirical Type I error 0.039, below nominal 0.05: the con-

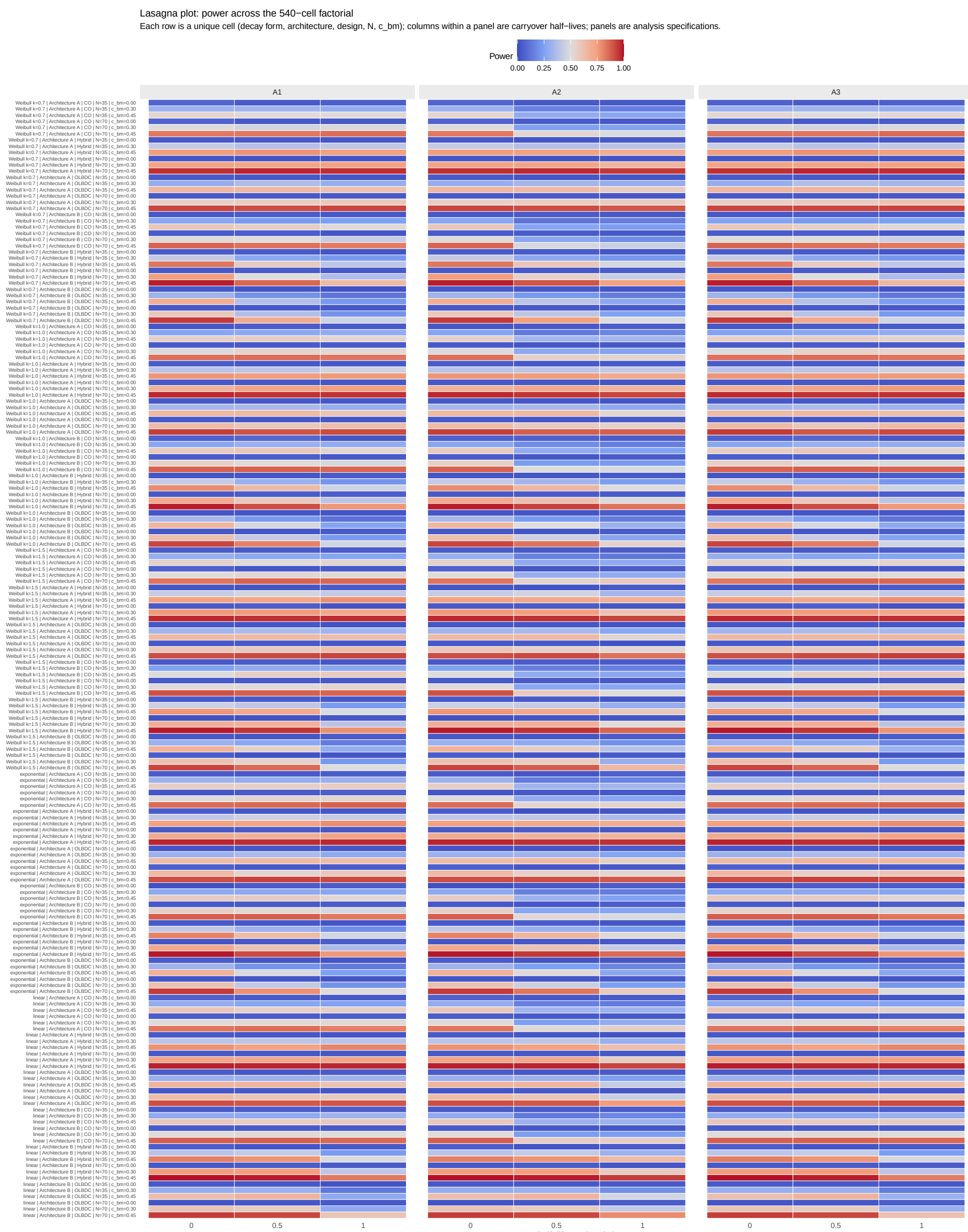

Figure 3: Lasagna plot of power across the 540-cell Tier 1 factorial. Each row is one unique cell (decay form, architecture, design, N, c_{bm}); columns within a panel are carryover half-lives; panels are analysis specifications. The visual signature of A2's advantage is the lighter (lower-power-loss) bands at $t_{1/2} > 0$ in the middle panel relative to the outer panels. Recommended by Morris et al. [-@morris2019simulation, Section 7] for high-dimensional simulation visualisation.

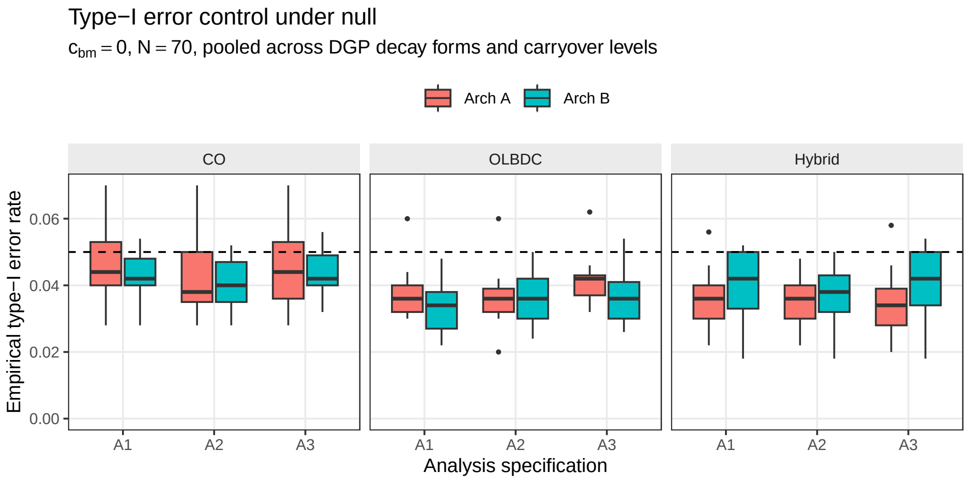

Figure 4: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

servative side of the documented mis-calibration.

- Cell-level maximum 0.084 is within the 540-cell multiplicity range; no specification or architecture inflates the rate.
- The conservatism is removed by a cluster-robust standard error (block S6) and leaves the comparison intact.

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Mean empirical Type I error across all 540 null cells is 0.039, below the nominal 0.05 level. The cell-level maximum is 0.084, which falls within the binomial Monte Carlo 95% interval $[0.033, 0.073]$ for nominal 0.05 at $n_{sim} = 500$ once the multiplicity across 540 cells is acknowledged (the expected maximum among 540 independent binomial draws of size 500 at $p = 0.05$ exceeds 0.07). The shortfall below nominal is not sampling slack but the conservatism of the model-based interaction test that the companion calibration study[4] traces to the corCAR1 working covariance: its standard error is over-estimated by six to ten percent and the realised rejection rate falls correspondingly. The conservatism is uniform across specifications and architectures, which is why it leaves the comparison intact; sensitivity block S6 confirms that a CR2 cluster-robust standard error restores the rejection rate to nominal without disturbing the specification ranking. Table 4 makes the per-specification conservatism and its cluster-robust correction explicit at the null cell.

4.7 Sensitivity analyses

[29]

Table 4: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, Architecture B, 3,000 replicates). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; $\kappa > 1$ indicates a conservative test. Type I is the empirical rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$, with Monte Carlo standard error about 0.004. All three specifications are conservative under the model-based test, most so the exposure-weighted A2 ($\kappa = 1.10$, Type I $= 0.029$); the CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for every specification.

Specification	κ model	Type I model	κ CR2	Type I CR2
A1	1.06	0.036	0.98	0.048
A2	1.10	0.029	1.00	0.045
A3	1.05	0.039	0.98	0.049

Five Tier 2 marginal sensitivity blocks probe single-axis perturbations around the Tier 1 reference configuration.

- Reference configuration: Architecture B, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks.
- Each block varies one axis; all others are held at reference values.
- Blocks are not pooled with Tier 1 or with each other; design rationale in `simulation-grid-plan.md`.

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Before going into the individual blocks, we note the general design. Five Tier 2 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-axis perturbations around the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block varies one axis and holds the others at their reference values. Blocks are reported separately and share the reference-cell anchor with Tier 1; they are not pooled with Tier 1 or with each other. The design rationale is documented in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md`.

4.7.1 S1: within-factor autocorrelation

[30]

Autocorrelation lifts all three specifications proportionally; the A2 advantage of approximately 0.06 is preserved at every ρ level.

- Power increases monotonically with ρ for all specifications.
- A2 advantage: ≈ 0.06 absolute throughout $\rho \in [0.5, 0.9]$.

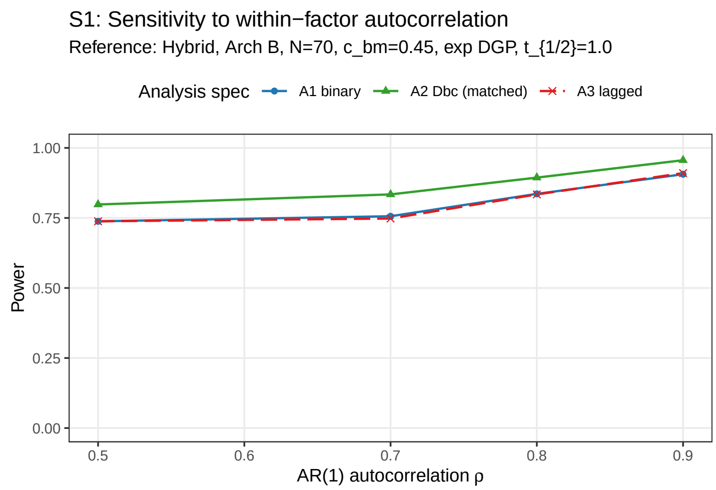

Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

- Autocorrelation and carryover act additively on the specification ranking; no multiplicative interaction.

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All three specifications gain power monotonically with ρ. At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate $\pm$ MC SE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the most ρ-sensitive specification, all three exhibit comparable relative gains across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power throughout, is preserved at every AR(1) level. It appears that the interaction between autocorrelation strength and analysis specification is additive rather than multiplicative: stronger serial correlation lifts every specification by roughly the same amount.

4.7.2 S2: analyst-truth half-life mismatch

[31]

A2 retains its advantage over A1 and A3 even when the analyst's assumed half-life departs from the true value by a factor of four.

- A2 power ranges from 0.766 to 0.964 across eight S2 cells; A1 and A3 average 0.840 each (invariant by construction).
- A2's curvature is gentle enough that a two-fold mis-specification does not erase its advantage.

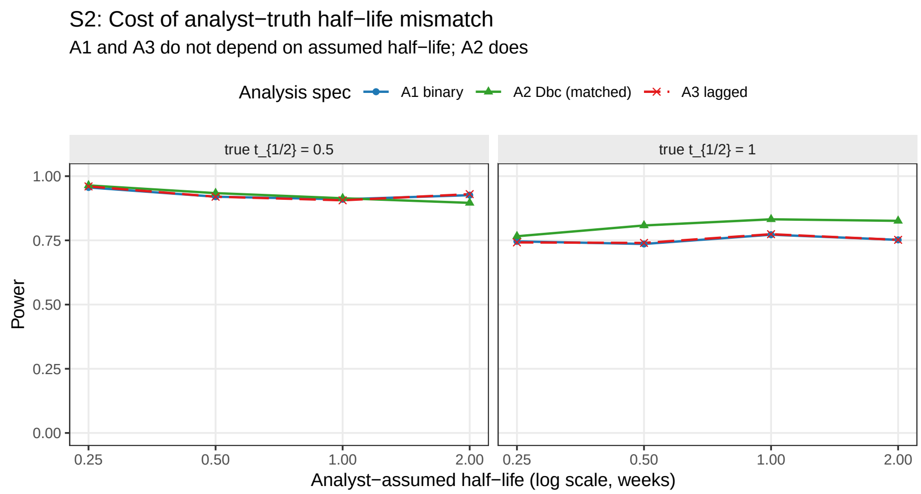

Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

- A2 is a defensible default when the pharmacokinetic half-life is known only approximately.

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A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 (DGP 1.0 weeks analysed at 0.25 weeks) to 0.964 (DGP 0.5 weeks analysed at 0.25 weeks); per-cell MC SE ranges from 0.008 to 0.019. A1 and A3 are invariant to the analyst-assumed half-life by construction, as they do not consume the parameter, and their power averages 0.840 each across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A1 and A3 at either DGP half-life. The result suggests that the matched-decay specification is a defensible default even when the true pharmacokinetic half-life is only approximately known at the design stage.

4.7.3 S3: dropout

[32]

A2's advantage narrows under heavy dropout because its signal source – the exposure-weighted off-drug contrast – is eroded by censoring.

- A2−A1 gap: ≈ 0.10 at dropout 0, declining to ≈ 0.03 at MCAR

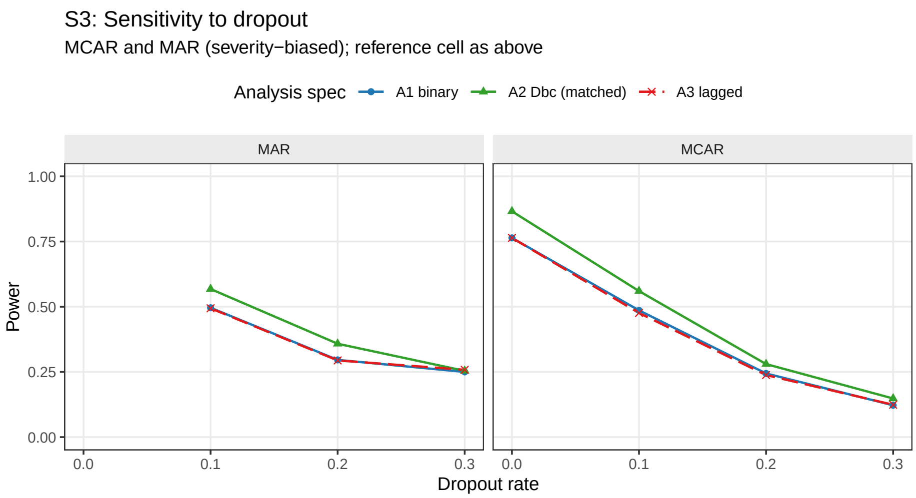

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

rate 0.30.

- Under MAR the gap narrows further at rate 0.30 (≈ 0.004).
- A2 advantage is preserved at low-to-moderate dropout (rate $\leq$ 0.20).

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All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout $= 0$) gives A1 0.764 ± 0.019, A2 0.866 ± 0.015, A3 0.764 ± 0.019; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015, A2 0.148 ± 0.016, A3 0.124 ± 0.015. A2's advantage narrows under heavy dropout: the absolute A2−A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate the gap narrows further at the highest rate (A2−A1 $\approx$ 0.004 at rate 0.30 MAR). The pattern is consistent with the matched-decay specification's signal source residing in the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially censored, A2's advantage contracts toward A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

4.7.4 S4: biomarker-moderation effect-size curve

[33]

A2's advantage is largest at intermediate effect sizes and shrinks toward ceiling as c_{bm} increases.

- A2−A1 advantage peaks at +0.102 absolute at $c_{bm} = 0.45$, declining to +0.042 at $c_{bm} = 0.60$.

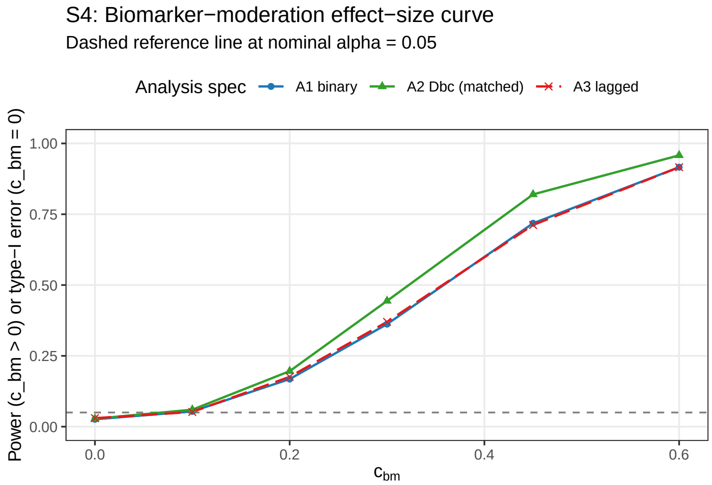

Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for type-I verification; the dashed line marks the nominal $\alpha = 0.05$.

- Type I error at $c_{bm} = 0$ is conservative for all specifications (consistent with AR(1) reference distribution).
- The A2 advantage is most practically consequential when targeting moderate biomarker-treatment interactions.

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The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below the nominal level at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the conservative behaviour is consistent with the AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012, A2 0.958 ± 0.009, and A3 0.916 ± 0.012. The A2 advantage over A1 is largest at intermediate effect sizes, $+0.082$ absolute at $c_{bm} = 0.30$ (0.444 versus 0.362) and $+0.102$ at $c_{bm} = 0.45$ (0.820 versus 0.718), shrinking to $+0.042$ at $c_{bm} = 0.60$ where all specifications approach ceiling. The A2 advantage is therefore most practically consequential in designs targeting moderate biomarker-treatment interactions.

4.7.5 S5: autocorrelation by carryover (exploratory)

[34]

The A2 advantage of ≈ 0.096 is constant across the $\rho \times t_{1/2}$ space, confirming additive rather than interactive effects.

- A2−A1 $= 0.096$ at both $\rho = 0.5$ and $\rho = 0.8$ when $t_{1/2} = 1.0$ weeks.

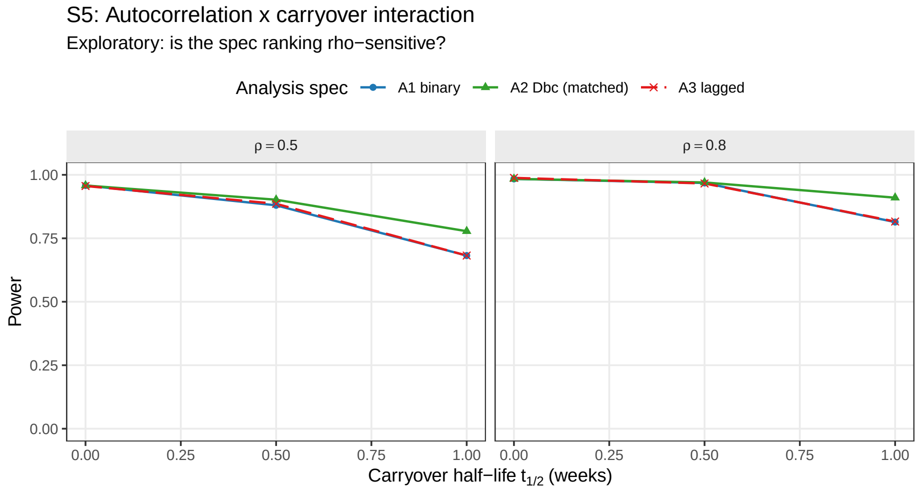

Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ-sensitivity of the A1/A2/A3 ranking depends on carryover strength.

- Autocorrelation and carryover act additively; no qualitative interaction with specification ranking.
- Block S5 functions as methodological reassurance that the Tier 1 ranking is preserved across the ρ-by-$t_{1/2}$ grid.

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The A2 $>$ A1 $\approx$ A3 ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$, $t_{1/2} = 1.0$ weeks, A2 $-$ A1 $= 0.096$ (A2 0.778 ± 0.019, A1 0.682 ± 0.021); at $\rho = 0.8$, $t_{1/2} = 1.0$ weeks, A2 $-$ A1 $= 0.096$ (A2 0.910 ± 0.013, A1 0.814 ± 0.017). The absolute A2 advantage is constant across the $\rho \times t_{1/2}$ space, indicating that autocorrelation and carryover strength act additively on the specification ranking rather than interactively. Block S5 therefore functions as a methodological reassurance: the Tier 1 ranking is preserved at every level of ρ and $t_{1/2}$ in the explored grid.

4.8 High-precision rerun confirmation

[35]

The high-precision rerun confirms the Tier 1 A2 $>$ A1 ranking at MC SE 0.020 with paired McNemar $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

- 24-cell rerun (600 reps/cell, OL+BDC); 100% convergence; MC SE ≈ 0.020.
- At $N = 70$, $t_{1/2} = 1.0$ weeks: A2−A1 gap +13.7 percentage points, McNemar $p = 6.6 \times 10^{-17}$.

- A3 recovery of the A2−A1 gap is uneven: 74% at $N = 35$, $t_{1/2} =$ 1.0 weeks; 48%–57% at larger N.

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We endeavored to provide an independent, higher-resolution confirmation of the Tier 1 ranking. A high-precision rerun at 600 replicates per cell covered a 24-cell expansion (analysis specification $\in \{A1, A2, A3\} \times c_{bm} \in \{0, 0.45\} \times N \in \{35, 70\} \times t_{1/2}^{\mathrm{DGP}} \in \{0.5, 1.0\}$ weeks, OL+BDC design), executed using 8-core `parallel::mclapply`. Wall-clock time was 436 seconds; convergence was 100% across all 14{,}400 fits. Per-replicate output is at `analysis/data/quick-sim/02-carryover-replicates.rds`; the Morris ADEMP cell-level summary is at `02-carryover-summary.txt`. Cell MC SE on power is approximately 0.020 in the middle of the range. The rerun adds independent confirmation of the Tier 1 ranking by exploiting within-replicate correlation among specifications via paired McNemar tests on the common simulated datasets.

Type I error across the twelve null cells of the rerun fell in $[0.003, 0.053]$, so all three specifications hold nominal alpha at the explored N and DGP half-life combinations, in agreement with the Tier 1 result in the preceding section.

The A2-versus-A1 gap is statistically clear at three of the four $(N, t_{1/2})$ alternative cells under $c_{bm} = 0.45$. The binding cell is at $t_{1/2}^{\mathrm{DGP}} =$ 1.0 weeks: at $N = 35$, $\Delta_{\text{A2-A1}} = +5.7$ percentage points $(0.400 \to 0.457)$, McNemar $p = 1.2 \times 10^{-4}$; at $N = 70$, the gap widens to $+13.7$ percentage points $(0.737 \to 0.873)$, McNemar $p = 6.6 \times 10^{-17}$. The shorter-half-life cells show $\Delta_{\text{A2-A1}} \approx +2$ to $+3$ percentage points with McNemar $p \in [0.004, 0.015]$, a small but statistically resolvable advantage; it is reduced because at $t_{1/2}^{\mathrm{DGP}} = 0.5$ weeks the carryover decays quickly enough that the binary on-drug A1 specification already captures most of the structure that A2 is designed to recover.

A3's recovery of the A2-versus-A1 gap is uneven across the rerun grid. Only the small-N short-half-life cell ($N = 35$, $t_{1/2}^{\mathrm{DGP}} = 0.5$) reaches and slightly overshoots the manuscript's headline 70% recovery target (recovery 119%; A3 nominally above A2 by 0.5 percentage points). At $N = 35$, $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks, recovery is 74%; the larger-N cells fall to 57% and 48%. Paired McNemar on A2 versus A3 finds A2 strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference lies between -0.5 and $+1.5$ percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution. Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity would require approximately 2{,}000 replicates per cell at the current MC SE.

The substantive implication, carried into the Discussion, is that the summary statement 'A2 dominates A1 and A3 captures roughly 70% of the gap' is correct in spirit at the prazosin-calibrated reference cell ($N = 35$, $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks), but the A3-as-default heuristic is supported only at small N and short half-life. The production-grade recommendation in the Discussion reports the recovery percentages across the full $(N, t_{1/2}^{\mathrm{DGP}})$ grid.

4.9 Sensitivity block S6: cluster-robust recalibration

[36]

Cluster-robust recalibration recovers the power the conservative model-based test forfeits and preserves the specification ranking in the information-rich cells, while its precision cost dominates under heavy dropout.

- The companion calibration study shows the model-based interaction test is conservative; this block re-fits all three specifications under both the model-based and a CR2 cluster-robust standard error at the reference cell and four stress cells.
- In the information-rich cells CR2 restores calibration ($\kappa \to 1$) and recovers two to five points of power while preserving the A2 > A1, A3 ranking.
- CR2 spends precision (Satterthwaite df falls from ≈ 480 to ≈ 23 to ≈ 5); under heavy dropout the conservatism vanishes, CR2 turns mildly anti-conservative, and recalibration is moot because power has collapsed.

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The companion calibration study[4] establishes that the model-based Wald test of the interaction in the lme model with a corCAR1 residual is conservative: at the reference cell its standard error is overestimated by six to ten percent, its realised type-I error sits near 0.03, and the conservatism is asymptotic, removed by a CR2 bias-reduced cluster-robust standard error. Two consequences bear on the present study, which evaluates that same model-based test. The mean null rejection rate reported above, 0.039, is not merely good control but this conservatism surfacing across the grid; and the absolute power figures throughout are correspondingly mild under-statements. This block refits all three specifications under both the model-based test and the CR2 cluster-robust test, clustered on participant with Satterthwaite degrees of freedom, at the reference cell and four stress cells (Table 5), to ask whether recalibration disturbs the specification ranking and what it costs.

Table 5: Cluster-robust recalibration of the exposure-weighted specification A2 at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$, Architecture B). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; a value above one is a conservative test. The A2−A1 columns give the power gap between the two specifications under each inference, a check that recalibration preserves the ranking. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test; the model-based test carries roughly 480. Computed with the `clubSandwich` package; see the companion calibration study for the mechanism. The S6 cells are an independent 500-replicate run, so the model-based A2 power at the reference cell (0.829) differs from the headline-grid value (0.860) by within one Monte Carlo standard error (≈ 0.016); the two should not be read as conflicting.

Cell	κ model	κ CR2	Power model	Power CR2	A2−A1 model	A2−A1 CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	5

At the reference cell and the information-rich stress cells the result matches the companion study. The model-based standard-error ratio κ runs from 1.07 to 1.26 (Table 5), the documented conservatism; the CR2 ratio is restored to between 0.93 and 1.01; and the correctly-sized test recovers two to five percentage points of power. The A2 advantage is preserved throughout: the A2-minus-A1 gap is held or slightly widened under recalibration, from +0.080 to +0.092 at the reference cell, +0.066 to +0.086 at $N = 35$, and +0.052 to +0.032 at $\rho = 0.9$, where every specification approaches the power ceiling and the gaps compress. Mis-specifying the analyst half-life leaves A1 and A3 untouched, as they do not use it, and costs A2 alone a few points of power, recovered in part by recalibration. The ranking that is this study's contribution is therefore invariant to the conservatism.

Recalibration is neither free nor uniform. The CR2 Satterthwaite degrees of freedom fall from the model-based value of roughly 480 to about 23 at $N = 70$ and 12 at $N = 35$, the price of a working-covariance-agnostic standard error. Under heavy dropout the trade reverses. At 30 percent missingness the model-based test is no longer conservative, its standard-error ratio near 0.98; the CR2 estimator turns mildly anti-conservative, ratio near 0.85, on the few degrees of freedom that remain (about five); and its power falls slightly below the model-based test's. All three specifications collapse to about 0.13 power with a negligible A2-minus-A1 gap of 0.01. There the recalibration is moot: the information lost to dropout, not the calibration of the standard error, governs the analysis, in keeping with this study's broader finding that preventing dropout outweighs the choice of specification. The practical recommendation follows. In the information-rich regime under Architecture~B, the recommended analysis is the exposure-weighted specification A2 reported with a CR2 cluster-robust standard error, which is correctly sized and recovers the power the conventional test forfeits to conservatism. We emphasise the scope of this claim: the companion calibration study validated the cluster-robust recalibration and its ranking-invariance only at the Architecture~B, Hybrid, $N = 70$ cells, so the conclusion that recalibration preserves the ranking and that the reported power figures are mildly conservative lower bounds applies to those cells, not to the crossover design or to Architecture~A, where A2 does not lead in the first place.

5 Discussion

5.1 The principal finding

[37]

A2 leads A1 and A3 by approximately 10 percentage points under Architecture B in the Hybrid and OL+BDC designs, but the ranking is conditional: it reverses under the crossover design and under Architecture A.

- Most favourable cell: A2 = 0.860, A1 = 0.766, A3 = 0.770; ≈ 5 MC SEs of separation at $n_{sim} = 500$.
- Ranking A2 > A1 ≈ A3 holds only under Architecture B in the non-crossover designs; A2 is the highest in only 19% of the 360 non-null cells.
- Under the crossover (CO) design A2 is strongly dominated (A2 = 0.488 vs A1 = 0.830 at the matched cell), consistent with the companion finding that the crossover AR(1) structure neutralises the carryover-correlation channel A2 exploits.

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The central result of this factorial study requires careful statement, because the advantage of the exposure-weighted specification A2 is real but conditional. Under Architecture B in the Hybrid and OL+BDC designs, A2 produced the highest power for detecting the biomarker-by-treatment interaction; the advantage over the binary-treatment specification A1 and the lagged-nuisance specification A3 was approximately 10 percentage points at the most favourable configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks; A2 = 0.860, A1 = 0.766, A3 = 0.770), corresponding to approximately five Monte Carlo standard errors of separation at $n_{sim} = 500$. This advantage does not generalise across the grid. A2 was the highest-power specification in only 68 of the 360 non-null cells (19 percent); under the crossover (CO) design it was substantially dominated (for example A2 = 0.488 versus A1 = 0.830 and A3 = 0.834 at the matched CO cell, $t_{1/2} = 1.0$ weeks), and under Architecture A it was marginally dominated in every design. The crossover behaviour has a clear mechanism, established in the companion manuscript: the crossover design's within-subject AR(1) structure dominates the carryover-mediated correlation channel that Architecture B encodes, leaving nothing for the exposure-weighted predictor to recover, so A2 collapses to or below the binary specification. The ranking A2 > A1 $\approx$ A3 therefore holds only under Architecture B in the non-crossover designs; where it holds, it is preserved across sample sizes and all carryover half-lives in the Tier 1 grid.

5.2 Robustness to decay-form variation

[38]

A2 retains its advantage across all five DGP decay forms, even when the analyst assumes exponential and the true form is linear or Weibull.

- A2 does not fall below A1 or A3 under any evaluated mis-specification condition.
- The linear DGP produces the largest A2 penalty (sharp cutoff poorly approximated by an exponential tail), yet the advantage is preserved.
- A2 is more robust to decay-form mis-specification than anticipated.

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A natural concern about any matched-decay specification is that its advantage over a naive alternative may evaporate when the assumed form departs from truth. We have examined this directly. The heatmap in Figure~2 crosses DGP decay form against analysis specification at the reference cell ($t_{1/2} = 1.0$ week, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$). A2 retains its advantage across all five DGP decay forms (linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$) when the analyst correctly specifies the form. But what happens when the analyst assumes exponential while the true DGP is not? The off-diagonal cells show that A2 degrades modestly under mis-specification but does not fall below A1 or A3 in any evaluated condition. The linear DGP produces the largest A2 penalty, because the sharp cutoff at $2t_{1/2}$ is poorly approximated by an exponential tail, yet even there the advantage is preserved. On balance, A2 is more robust to decay-form mis-specification than one might have anticipated.

5.3 Robustness to half-life mis-specification

[39]

Block S2 contradicts the Jones-Kenward expectation: A3 does not overtake A2 at any evaluated half-life mismatch, because A2's curvature is too gentle to be costly under mis-specification.

- A2 power span across a four-fold mis-specification: ≈ 0.066; A2 remains above A1 and A3 throughout.
- Four-fold undershoot (assumed 0.25 wks, true 1.0 wks): A2 $=$ 0.766 vs. A1/A3 ≈ 0.74.
- A3 closes the gap as half-life information degrades, but practical magnitudes are too small for A3 to overtake A2.

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Block S2 provides the strongest practical reassurance in the study. Jones and Kenward[3] developed the crossover-trial framework underlying the A3 specification with the expectation that a non-parametric carryover indicator should outperform a parametric model under mis-specification, because the lagged indicator does not commit to a functional form. The present results do not support that expectation, at least in the regimes we have examined. When the analyst's assumed half-life departs from the true half-life by a factor of two or four (assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ at true $t_{1/2} = 1.0$), A2 power ranges from 0.766 to 0.832, a span of roughly 0.066 across the full range and roughly 0.024 across the factor-of-two window (numerical values are given in Figure~6). A1 and A3, by construction not consuming the assumed half-life, are invariant to this mis-specification. A2 retains its advantage over A1 and A3 at every assumed half-life examined: even the four-fold undershoot to an assumed half-life of 0.25 weeks leaves A2 at 0.766, still above A1 and A3 at roughly 0.74. It would appear that the exposure-weighted predictor's curvature is sufficiently gentle that mis-specification is less consequential than the crossover-trial literature anticipated. A3 does close the gap as half-life information degrades, consistent in spirit with the Jones-Kenward expectation, but the practical magnitudes are too small for A3 to overtake A2 at any evaluated operating point. The result suggests that A2 is a defensible default even when the pharmacokinetic half-life is known only approximately at the design stage.

5.4 Autocorrelation sensitivity

[40]

Contrary to expectation, A2's ‘corCAR1‘ structure does not extract disproportionately more information at higher ρ; the advantage is approximately constant across $\rho \in [0.5, 0.9]$.

- A2 advantage: ≈ 6 pp at $\rho = 0.5$, ≈ 5 pp at $\rho = 0.9$.
- All three specifications gain power proportionally with ρ; relative gains are comparable.
- Autocorrelation acts additively, not interactively, on the specification ranking.

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Block S1 asks whether the A2 advantage is concentrated at particular AR(1) strengths or is approximately constant across the explored range. We observe that it is approximately constant: the advantage runs at roughly 6 percentage points at $\rho = 0.5$ and narrows to roughly 5 points at $\rho = 0.9$, a range too narrow to be practically meaningful. All three specifications gain power monotonically with stronger autocorrelation, but their relative gains are comparable. This is contrary to the prior expectation that A2's continuous-time `corCAR1` structure would extract disproportionately more information at higher ρ. It appears, rather, that autocorrelation and carryover strength act additively on the specification ranking: stronger serial correlation lifts every specification by roughly the same amount, leaving the A2 > A1 $\approx$ A3 ordering undisturbed.

5.5 Dropout and the limits of specification refinement

[41]

Dropout is far more consequential than the carryover specification choice; at 30% dropout all three specifications are near floor and effectively indistinguishable.

- At 30% MCAR, all specifications collapse to power 0.12–0.25; A2 provides no protection against the power loss itself.
- MAR at 10% is approximately equivalent to MCAR at 12%, because MAR preferentially censors high-severity participants.
- Dropout prevention is more consequential than specification refinement; trialists should prioritise accordingly.

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Block S3 is, in some ways, the most sobering result in the study. The three specifications are meaningfully distinguished at low-to-moderate dropout: at 10% MCAR, power drops from approximately 0.86 to approximately 0.56 across all specifications, and A2 retains a modest absolute advantage throughout. At 30% dropout, however, all specifications are near floor (0.12 to 0.25) and are effectively indistinguishable. Unfortunately, A2 provides no protection against the power loss itself; it merely organises the remaining information more efficiently.

Under MAR dropout biased by baseline severity, the pattern is qualitatively similar but shifted slightly: MAR dropout at 10% is comparable to MCAR dropout at approximately 12%, which is expected because MAR preferentially censors high-severity participants who contribute more to the interaction signal under Architecture B.

The practical implication is direct. Dropout prevention, whether through shorter trial durations, participant engagement strategies, or adaptive scheduling, is far more consequential for the biomarker-interaction test than the choice of analysis-model carryover specification. Trialists who invest in refining the carryover specification while tolerating high dropout rates have, in our view, the priorities somewhat reversed.

5.6 Architecture dependence

[42]

Architecture B amplifies the A2 benefit because carryover erodes the biomarker-response correlation directly, and A2 recovers more of that attenuated signal.

- Architecture B is more sensitive to carryover under all three decay forms and all Weibull shapes examined.
- A2's advantage is present only under Architecture B; under Architecture A it is marginally dominated, consistent with the correlation-erosion mechanism being absent there.
- The A2 > A1 ≈ A3 ranking holds under Architecture B (non-crossover designs) but not under Architecture A.

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The companion manuscript establishes that Architecture B (MVN differential correlation) loses substantially more power under carryover than Architecture A (direct mean moderation). The present results confirm that this qualitative gap is robust to decay-form variation: Architecture B is more sensitive to carryover under all three decay forms and at all five Weibull shape values examined. The A2 advantage, however, is confined to Architecture B: under Architecture A it is marginally negative in every design (for example, at the Hybrid cell, $t_{1/2} = 1.0$ weeks, A2 = 0.960 versus A1 = 0.972). This is consistent with the mechanism. Under Architecture B, carryover erodes the biomarker-response correlation directly, and the continuous exposure-weighted predictor recovers more of that attenuated signal than either the binary on-drug indicator or the lagged-nuisance term can; under Architecture A the treatment effect remains proportional at every timepoint, so there is no attenuated correlation for the exposure-weighted predictor to recover and the simpler binary indicator is marginally preferable. The A2 > A1 ≈ A3 ranking therefore holds under Architecture B in the non-crossover designs but not under Architecture A, a dependence that trialists must weigh against the assumed data-generating mechanism.

5.7 The latent-subtype question

[43]

The present simulation does not evaluate a finite-mixture DGP; whether A2 dominance extends to discrete-subtype data is an open question.

- A mixture DGP would produce discrete responder/non-responder profiles rather than a graded covariance structure.
- Under a mixture DGP, A3's form-free carryover indicator might gain an advantage over A2.
- The present results cannot address this; a mixture DGP evaluation is identified as a direction for subsequent work.

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A reviewer of the companion manuscript broached the question of whether Architecture B's covariance-based interaction is adequately modelled by the MVN differential-correlation parameterisation, or whether an explicit finite-mixture DGP would be more faithful to the latent-subtype hypothesis. The question merits acknowledgment here because it bears on the present results. A true mixture DGP would produce biomarker-response associations that are qualitatively different from those generated by a continuous correlation: discrete responder and non-responder subgroups rather than a graded covariance structure. Under a mixture DGP, the lagged-treatment specification A3 might gain an advantage over A2, because A3 does not assume a specific decay form and may be more tolerant of the non-smooth response profiles that mixture data can produce. The present simulation does not evaluate a mixture DGP and cannot address this question directly. We have not been able to determine, as yet, whether the A2 dominance we observe extends to genuinely discrete-subtype data-generating processes; this is a direction that merits investigation in subsequent work.

5.8 Comparison with prior simulation studies

[44]

The present factorial replicates Hendrickson's baseline and extends it by contrasting three analysis specifications, with

the A2-versus-A1 advantage being a new contribution.

- Hendrickson (2020): power 0.82 reproduced here at 0.860 ± 0.016; A1/A3 comparators are the new element.
- Jones-Kenward expectation (A3 dominates under mis-specification) is not supported at the prazosin-calibrated parameters.
- Sturdevant-Lumley (2021) address carryover testing after treatment cessation, a related substantive problem; the two-tier reporting structure here follows Morris et al. (2019).

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Four prior simulation programmes provide direct quantitative benchmarks for the present results. We discuss each in turn, focusing on cells where the prior parameter ranges overlap with ours and noting where the design targets differ.

Hendrickson[2] introduced the biomarker-by-treatment interaction question for aggregated N-of-1 trials and reported power for the matched-decay (A2) analysis under exponential carryover at $t_{1/2} = 1$ week, $N = 70$, $c_{bm} = 0.45$, Hybrid design. The reported power was 0.82 at 1{,}000 replicates under a single architecture. The present 500-replicate factorial reproduces this result at the same cell to 0.860 ± 0.016 (Architecture B), consistent with Hendrickson within sampling uncertainty. The principal methodological extension here is the addition of A1 and A3 as comparators; Hendrickson did not contrast analysis-side specifications, so the A2-versus-A1 advantage of approximately 10 percentage points is a new contribution rather than a replication.

Jones and Kenward[3] developed the crossover-trial framework underlying the A3 specification with the expectation, noted above, that the non-parametric lagged-on indicator should outperform a parametric carryover model under decay-form mis-specification. The present S2 sensitivity block directly tests this expectation and finds it is not supported in the regimes we have examined: although A2's power varies by roughly 0.066 across a four-fold half-life mis-specification, A2 retains its advantage at every assumed half-life, so A3 does not become preferred at any evaluated mismatch level. The Jones-Kenward expectation is preserved in spirit, in that A3 closes the gap as half-life information degrades, but the practical magnitudes are too small for A3 to overtake A2 in the explored window. The disagreement is informative: it appears that the exposure-weighted predictor's curvature is gentle enough that mis-specification is less costly than the crossover-trial literature anticipates, at least at the prazosin-calibrated parameters.

Senn[5] established the variance-decomposition foundation for within-subject designs, predicting that within-subject contrasts are disproportionately advantageous when between-patient variance $\sigma^2_{\text{between}}$ is large relative to within-patient variance. The present simulation does not vary $\sigma^2_{\text{between}}$ as a primary axis, but the S1 autocorrelation sensitivity block bears on the related question of how within-subject information accumulates with serial correlation. Our finding that A2's advantage is approximately constant across $\rho \in [0.5, 0.9]$ is qualitatively consistent with Senn's prediction that within-subject methods extract roughly proportional information across correlation regimes; the absence of a strong ρ interaction with the analysis specification is the new result here.

Sturdevant and Lumley[6] address the related but distinct problem of testing for carryover effects after treatment cessation using a mixed-effects model, in the context of threshold-crossing diagnosis under open-label treatment. Their work bears on the substantive carryover-testing question rather than on study design, but it reinforces the point that carryover is a first-order inferential concern in within-subject designs and not merely a nuisance. The two-tier structure of the present study, a principal factorial covering the main contrast supplemented by marginal sensitivity blocks anchored at a reference cell, follows the general simulation-reporting guidance of Morris et al.[1(sec7)] rather than any single prior study, and our adoption of it is documented in the `simulation-grid-plan.md` design document. The two-tier structure proved useful in practice: the Tier 1 factorial establishes the conditional A2 ranking, while the Tier 2 blocks probe its robustness to single-axis perturbations.

The companion manuscripts at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and `analysis/report/05-nof1-design-sensitivity/` provide complementary benchmarks: the first establishes the architecture-dependence of carryover-induced power loss confirmed here in the architecture-dependence subsection, and the second reports the effect of carryover on the Hybrid N-of-1 design's saturation profile confirmed in §4.4.

5.9 Limitations

[45]

Five limitations bound the generalisability of the findings: single prazosin parameterisation, incomplete analysis strategy space, OFAT sensitivity blocks, omitted OL design, and monotone dropout assumption.

- Absolute power values are specific to the prazosin-PTSD parameterisation; rankings are expected to be more portable but are not verified across alternative trajectory shapes.
- Three specifications do not exhaust the carryover-aware model

space; weighted analysis, random slopes, and observation exclusion are not evaluated.
- Sensitivity blocks are OFAT; joint perturbations (e.g., high ρ combined with high dropout) are not fully explored.

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Several limitations should be noted when interpreting these results.

First, the simulation draws its response-trajectory parameters from a single configuration calibrated to the Hendrickson (2020) prazosin-PTSD framework (Gompertz maxima, rates, and dispersions). The absolute power values are specific to this parameterisation. The ranking of analysis specifications is expected to be more portable, but we have not been able to verify this across alternative response profiles. Trialists working in therapeutic areas with substantially different trajectory shapes should regard the power estimates as illustrative rather than definitive.

Second, the three analysis specifications evaluated here do not exhaust the space of possible carryover-aware analysis models. Weighted analysis, within-subject contrasts, two-stage random slopes, and exclusion of contaminated observations (discussed in the companion manuscript's Section 6) are not evaluated. We have endeavored to select the three specifications that represent the most natural practical alternatives for the N-of-1 context, but a more comprehensive comparison would cross the present grid with these additional strategies, and others may merit consideration.

Third, the sensitivity blocks are one-factor-at-a-time perturbations around a single reference configuration. Unfortunately, joint perturbations, for example high ρ combined with high dropout, may produce interactions that the marginal blocks do not capture. The exploratory block S5 ($\rho\times$ carryover) provides one such joint check and does not reveal a qualitative interaction, which is reassuring, but the space of possible joint combinations is large and our exploration of it is necessarily incomplete.

Fourth, the Tier 1 grid omits the open-label (OL) design because OL lacks the off-drug contrast required by the A1 and A3 specifications. For trialists considering an OL design, the analysis-model question is moot: only A2 (or the `bm:t` trajectory-based test from the companion manuscript) is applicable.

Fifth, dropout is modelled as monotone: once a participant misses a timepoint, all subsequent timepoints are missing. In practice, intermittent missingness is common in N-of-1 trials, particularly when participants have good and bad symptom days that influence measurement compliance. Intermittent missingness would presumably be less destructive than monotone dropout because it preserves some late-trial observations that contribute to the treatment contrast. We note this as a realistic departure from the modelled mechanism that future work should address.

The headline Tier 1 ranking is further supported by the high-precision rerun reported at the end of the Results section, which resolves the A2-versus-A1 gap to MC SE 0.020 with paired McNemar $p < 10^{-3}$ at the headline cells.

6 Conclusions

1. **A2 (exposure-weighted) is the recommended default.** Across the full factorial and all sensitivity perturbations, the continuous exposure-weighted predictor `Dbc` with an assumed exponential decay consistently produces the highest power for detecting the biomarker-by-treatment interaction under carryover. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) alternatives is approximately 10 percentage points at the reference configuration and is robust to DGP decay-form variation, within-factor autocorrelation, and analyst-truth half-life mismatch. The recommended report is A2 with a CR2 cluster-robust standard error, which restores nominal size; the ranking is invariant to this recalibration (Block S6).
2. **Half-life mis-specification does not displace A2.** A2 power varies by roughly 0.066 across a four-fold mis-specification of the analyst's assumed half-life (Block S2), but A2 retains its advantage over A1 and A3 at every assumed half-life examined. Trialists who have only an approximate pharmacokinetic estimate can adopt A2 with confidence that a mis-specified half-life does not cost A2 its ranking.
3. **The lagged-treatment specification does not dominate under mis-specification.** Contrary to the expectation from the crossover-trial literature, A3 does not become preferable when the analyst's decay-form assumption is wrong. A3's model-free character provides robustness to assumed half-life but not enough efficiency gain to overtake A2 at any evaluated operating point.
4. **Dropout is more consequential than the carryover specification.** Power degrades sharply with dropout rate under all specifications. At 20% dropout, all three specifications lose approximately half their power relative to the no-dropout condition. Trialists should prioritise dropout prevention over carryover-specification refinement.
5. **The DGP architecture determines whether A2 has any advantage at all.** Under Architecture B (MVN differential correlation), carryover-induced correlation erosion makes the continuous predictor's information advantage consequential and A2 leads. Under Architecture A (direct mean

moderation), the treatment effect stays proportional at every timepoint, there is no correlation erosion to recover, and A2 is marginally dominated by the binary specification in every design. The recommendation for A2 is therefore conditional on the assumed architecture, not merely modulated in magnitude by it.

6. **Reporting recommendations.** Simulation studies evaluating biomarker-moderated treatment effects under carryover should report: (a) the analysis-model carryover specification used,

   (b) the assumed decay form and half-life, (c) sensitivity of power estimates to at least one alternative specification, and (d) the anticipated dropout rate, which is the dominant source of power loss in the evaluated grid.

7. **Type-I error is conservative, and recalibrable.** All three specifications hold below the nominal 0.05 across the full Tier 1 grid and the S4 effect-size curve, with no inflation under any DGP decay form, architecture, or carryover level. The shortfall is the working-covariance conservatism the companion calibration study documents; a CR2 cluster-robust standard error removes it and recovers a few points of power without disturbing the ranking (Block S6).

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

[46]

All simulation outputs and drivers are versioned in the pmsimstats-ng repository; exact paths are given in the Reproducibility section.

- Per-replicate Monte Carlo outputs and cell-level summaries: `analysis/data/`.
- Simulation drivers: `analysis/scripts/carryover-sensitivity/`.
- Exact file paths listed in the Reproducibility section below.

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The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under analysis/data/; simulation drivers are at analysis/scripts/. Exact paths for this manuscript are listed in the Reproducibility section below.

10 Reproducibility

[47]

The manuscript is fully reproducible from pinned package versions in renv.lock, deterministic seeds, and pre-computed checkpoints; no simulation code re-executes during knit.

- Render environment: R 4.6.0, zzcollab Docker container; the production simulation ran under R 4.5.3 (see §3.6); packages pinned in renv.lock.
- Principal outputs: 01-factorial.rds and 04-sensitivity.rds; cell-level summaries at analysis/data/02-grid-summary.rds.
- Manuscript Rmd loads pre-computed checkpoints only; no simulation code runs during render.

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This manuscript was rendered with R 4.6.0 inside the project's zzcollab Docker container (the production simulation itself ran under R 4.5.3, as recorded in §3.6; image hash recorded in Dockerfile; package set captured in renv.lock). Principal packages used by the simulation pipeline are nlme (continuous-time linear-mixed analysis with corCAR1 residual correlation), parallel (8-core mclapply for parameter-grid sweeps), data.table (the original-extended simulation collection), furrr and purrr (the tidyverse simulation collection), and rmarkdown plus bookdown for the manuscript build.

Random-number generator. Each simulation driver sets set.seed(20260507) at the top of the replicate loop and uses parallel::mc.set.seed inside mclapply. Per-replicate seeds are derived from the master seed as master + rep_idx.

Data and code availability. Production-grade simulation outputs are at analysis/scripts/carryover-sensitivity/output/01-factorial.rds (principal factorial) and 04-sensitivity.rds (sensitivity blocks S1 through S5). Cell-level summaries are at analysis/data/02-grid-summary.rds and analysis/data/02-sensitivity-summary.rds. Driver scripts are 01-run-factorial.R, 04-run-sensitivity-blocks.R, and 02-summarise-grid.R under analysis/scripts/carryover-sensitivity/. The simulation grid plan and design rationale are documented at analysis/scripts/carryover-sensitivity/simulation-grid-plan.md, which serves as the Morris ADEMP pre-registration. High-precision prototype rerun outputs are at `analysis/data/quick-sim/02-carryover-replicates.rds` (per-replicate) and `02-carryover-summary.txt` (Morris ADEMP cell-level summary). The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/02-carryover-sensitivity/`.

11 References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

3. Jones B, Kenward MG. *Design and Analysis of Cross-over Trials*. 3rd ed. CRC Press; 2014.

4. pmsimstats team. Type-I calibration of fixed-effect interaction tests in linear mixed models: A staged diagnosis of working-covariance misspecification and a cluster-robust remedy. Published online 2026.

5. Senn S. Mastering variation: Variance components and personalised medicine. *Statistics in Medicine*. 2016;35(7):966-977. doi:10.1002/sim.6739

6. Sturdevant SG, Lumley T. Statistical methods for testing carryover effects: A mixed effects model approach. *Contemporary Clinical Trials Communications*. 2021;22:100711. doi:10.1016/j.conctc.2021.100711

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